

How disrupted spatial fluxes reshape food web structure along the river–sea continuum

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Abstract

Introduction

This is some text.

Connectivity

Key mechanisms

- Small dispersal stabilizes through spatial structure and asynchrony.
- Refuge effects occur when predator have lower efficiency when prey is rare (Holling type III).
- Prey switching of mobile top predators can stabilize connected food webs through ‘spatial coupling’.
- At high dispersal/connectivity the stabilizing effect wanes as strongly connected patches behave as a single patch.
- Studies between pairs of the following connectivity, complexity and stability are common. However, studies investigating how these are interconnected are rare.
- Connectivity as 4 dimensions: longitudinal, lateral, vertical and temporal. Grill et al. (2019)

Important references

- LeCraw 2014 - Review the effect of connectivity on food web complexity and stability.
- McCann 2005 - Consumers link food web in space, and their response can shape the spatial response of coupled food webs.
- Pillai 2011 - Metacommunity theory shows that food web complexity is maximized at intermediate level of colonization rates.

Dams

Key mechanisms

- Decrease river connectivity / reduce migration.
- Reservoir, lotic from lentic conditions, can result in hypoxic conditions. Favor phytoplanktons and macrophytes.
- Reservoir increase residence time of nutrients upstream and decrease the downstream output, which can result in a decrease in primary productivity.
- Hydropeaking - that is, the sudden release of large water to answer the higher electric demand - can harm riverine community due to sudden change in water level and water flow.
- Cumulative impact of dams can reduce the nutrient input in deltas.

Important references

- Ding 2026 - more endangered species near dam at a global scale.
- Dean 2023 - cumulative impact of downstream dam on potamodromous species.
- He 2024 - review dam impacts on riverine community.
- Pedersen 2012 - associate the drastic decline in the European eel to dam impacts.

LeCraw, Kratina, and Srivastava (2014)

Appendix

Model DAG

Put model DAG here using targets.

Grill, G., B. Lehner, M. Thieme, B. Geenen, D. Tickner, F. Antonelli, S. Babu, et al. 2019. "Mapping the World's Free-Flowing Rivers." *Nature* 569 (7755): 215–21. <https://doi.org/10.1038/s41586-019-1111-9>.

LeCraw, Robin M., Pavel Kratina, and Diane S. Srivastava. 2014. "Food Web Complexity and Stability Across Habitat Connectivity Gradients." *Oecologia* 176 (4): 903–15. <https://doi.org/10.1007/s00442-014-3083-7>.